

PAPER_952: 26-State HRes Spectral Ladder

Author: Daniel T. Murphy - Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 214 **Source:** spectral_ladder_26state.py (SpectralLadder26State) **Calculator:** SpectralLadder26StateCalc (CP4 #536) **CVW:** v2.0.0 compliant

Abstract

We derive the 26-state HRes (Hydrogen Resonance) spectral ladder, the energy progression across quantum layers from proto-H ($n = 1$) to proto-Fe ($n = 26$). The energy levels $E_n = E_0 \cdot (2\pi)^{n/3} \cdot S_{26}([SSq])$ form a geometrically growing ladder stabilizing the vacuum ratio $\rho_{SCm}/\rho_{UA} = 0.1$ and driving phonon resonance at every layer.

1. Energy Levels

$$E_n = E_0 \cdot (2\pi)^{n/3} \cdot S_{26}([SSq]), \quad n = 1, \dots, 26$$

where $E_0 = \hbar\omega_{SCm}$ and $\delta_n = (2\pi)^{n/6}$.

2. Element Mapping

n	Z_{id}	Element	Type
1	1	proto-H	Magnetic
14	14	proto-Si	Non-magnetic
26	26	proto-Fe	Magnetic

The ladder maps the complete magnetic/non-magnetic hierarchy from hydrogen through iron.

3. Source Data

- **File:** spectral_ladder_26state.py
 - **Session:** 214
 - **CP4 Class:** SpectralLadder26StateCalc (#536)
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References

1. Murphy, D.T. - Star Magic UQFF Framework (2024-2026)